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Multi-period Carbon Capture, Utilization, and Storage (CCUS) infrastructure design and planning under uncertainty

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ABSTRACT

The design of large-scale Carbon Capture, Utilization, and Storage (CCUS) networks is nontrivial due to large number of sources and sinks and their potential connectivity. The deployment of capture plants and CO_2 transportation pipelines is a gradual process with facilities coming online at different points in time. Uncertainties in underground storage volumes for CO_2 sequestration and enhanced oil recovery pose additional challenges. In this work, we employ a multi-period model formulation and a stochastic optimization approach combined with a rolling horizon-based solution strategy to find optimal CCUS network designs and infrastructure deployment schedules under a wide range of future realizations of the geological storage volumes. We also consider constraint on the budget allocation for each period. Our results for a case study on CCUS for the Illinois Basin region in the U.S. show that selecting capture facilities with high CO_2 emissions is both economically and environmentally favorable. Utilization provides a strong economic incentive and is favorable towards the early deployment of CCUS at sites that can generate revenue through enhanced oil recovery. Accounting for uncertainties in geological storage volumes enable recourse actions such as building additional pipeline connections to mitigate undesirable deviations from the planned capture levels. These indicate that incorporating uncertainties in CCUS network design and the planning of the deployment of such infrastructure is important to reduce deviations from the expected capture amount as compared to a deterministic approach.

1. Introduction

Carbon Capture, Utilization, and Storage (CCUS) could account for nearly one-sixth of the total cumulative reductions in CO_2 emissions (IEA, 2020) in the future. CCUS involves capturing CO_2 from stationary sources by an appropriate technology, such as chemical absorption or pressure swing adsorption, and transporting it via pipelines for long-term sequestration in underground geological formations. The captured CO_2 can also be utilized thorough CO_2-Enhanced Oil Recovery (CO_2-EOR). Fig. 1 depicts a schematic of the CCUS network design process.

With a large number of sources and sinks to be appropriately and optimally matched, the CCUS network design problem is combinatorially complex (Hasan et al., 2014). Several mathematical models have been proposed in the literature for CCUS network optimization. They focused on distinct aspects of CCUS network design and differ from each other in terms of scale and complexity. As an indicative example, SimCCS (Middleton and Bielicki, 2009) was developed to identify a CCS network topology and CO_2 pipeline sizing for California. For CCUS network optimization with simultaneous selection of materials and capture technologies (Hasan et al., 2014, 2015) introduced a multi-scale framework for nationwide and regional networks for the United States. Under different scenarios of sink storage capacities and injectivity and source lifetimes, Diamante et al. (2013) proposed a graphical approach for matching multiple CO_2 sources and storage sites. By discretizing the European territory for spatially explicit definitions of large emission clusters, d'Amore et al. (2018) used a MILP framework for the strategic design and planning of a large European supply chain for geological sequestration. Massol et al. (2018) explored the impacts of constructing several regional networks instead of a single grid for the Iberian peninsula. Wang et al. (2020) explored minimum cost source–sink relationships and the development of such source–sink clusters for China. For an extensive review of CCUS optimization models, we direct the reader to Zhang et al. (2022).

Moreover, the construction and deployment of infrastructure such as capture plants and CO_2 transportation pipelines for a CCUS network is a gradual process with such facilities coming online at different

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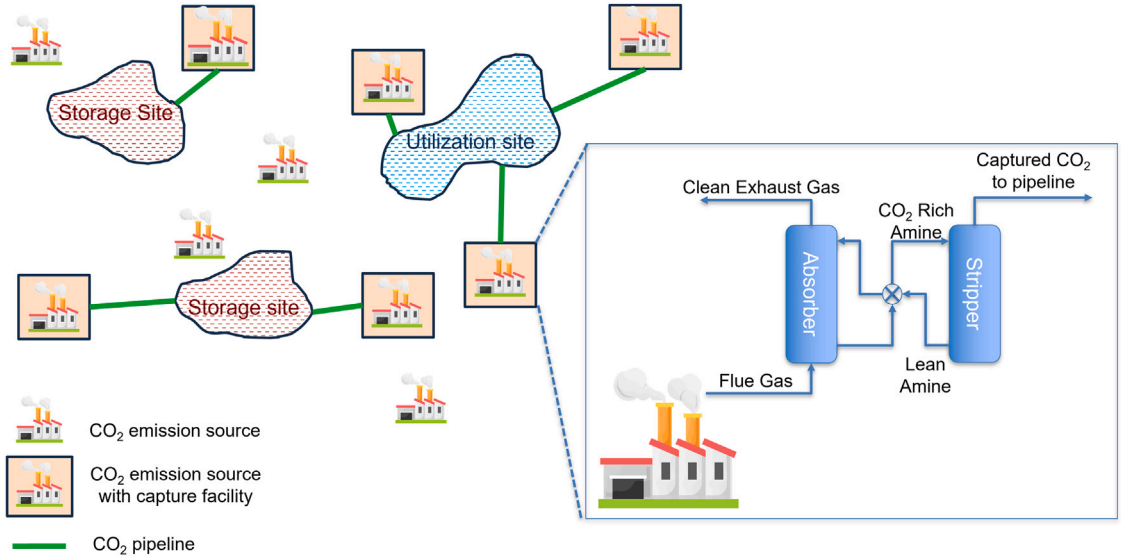


Fig. 1. Schematic diagram of a CCUS network. Designing a CCUS network entails capturing CO_2 using an appropriate technology such as amine-based absorption at selected CO_2 emission sources and transporting them using pipelines for injection into underground storage reservoirs or for CO_2-EOR.

points in time over a planning period. This necessitates the use of multi-period optimization models. Tan et al. (2012) developed a mixed integer nonlinear programming (MINLP) model, for continuous-time optimization for source–sink matching in carbon capture and storage systems and solved it as a reformulated mixed-integer linear program (MILP). For the multi-period deployment of CCUS infrastructure across Europe, InfraCCS (Morbee et al., 2012) incorporated the automatic clustering of sources and sinks. Middleton et al. (2012) extended the SimCCS to a multi-period setting to find potential CCS infrastructure deployment schedules for the Texas Panhandle area. Fraga et al. (2021) incorporated multiple modes of carbon capture and developed a multi-period MILP for the cost optimization of CCUS networks in Norway. Duarte et al. (2022) presented an MILP formulation for maximizing the net present value (NPV) of a CCUS supply chain considering different capture technologies and multi-modal transport options for the captured CO_2. Fraga et al. (2024) extended their previous multi-period model to include considerations for CO_2 conditioning, pipeline and batch-wise transportation, intermediate hub storage and injection at storage fields. Montagna et al. (2022) developed an MILP for the design and optimization of long term CCUS supply chain networks with a particular focus in a comprehensive modeling of the CO2-EOR operations.

Additionally, uncertainties in model parameters make it even more difficult to find a design and deployment schedule which is good enough or, in some cases, robust for different realizations of these uncertain parameters. Multiple previous studies have addressed how to deal with different uncertainties in a multi-period setting. For instance, Han and Lee (2012) developed a multi-period stochastic MILP for CCUS infrastructure deployment on the eastern coast of Korea under uncertain CO_2 emissions, product selling prices and plant operational costs. He et al. (2014) used a multi-period stochastic MILP framework for optimal source–sink matching considering uncertainties in the operating lives of sources, upper limits of the CO_2 storage capacities for the sinks, and the carbon footprint of the compensatory power to make up for energy losses. Li et al. (2022) developed a stochastic MILP framework that integrates the design of sustainable water, energy, waste treatment and CCUS networks to minimize CO_2 emissions and waste generated. Wang and Papadokonstantakis (2024) proposed a multi-period optimization framework for CCUS activities in the Austrian cement industry with uncertainties in production costs, carbon and product pricing, and storage capabilities. Tapia (2024) incorporated the management of CO_2 storage risk with the objective of maximizing satisfaction through minimizing such risks, and minimizing indeterminacy arising from the inaccuracy of the storage estimates. In our previous work (Aras et al., 2025), we studied the effect of geological storage volume and selling price uncertainties for designing CCUS networks, where we employed a two-stage optimization approach for the quantification of these uncertainties. However, to obtain a good deployment schedule, a multi-period stochastic optimization approach is required which is the focus of this work. Furthermore, limitations on financial resources which are required for the deployment and operation of capture plants and pipelines, necessitate the inclusion of periodic budget allocation constraints. Such economic constraints must be taken into account for finding an optimal planning schedule with the objective of minimizing the societal effects of CO_2 emissions.

The novel features of this work include: (1) a methodology for multi-period deployment of CCUS infrastructure over a deployment time horizon to minimize the effect of geological storage volume uncertainty; and (2) provision of an approach to gain insights for the strategic decision making under limited annual budget allocation in each time period.

This article is organized as follows. The CCUS network design problem statement is formally described in Section 2 and models for stochastic CCUS network design are introduced in Section 3. The methodology used to solve the problem is presented in Section 4. The results for multi-period CCUS infrastructure deployment in the Illinois Basin region are presented in Section 5, followed by concluding remarks in Section 6.

2. Problem statement

In this section, we define the scope of CCUS network infrastructure planning problem considered in this work. We state the problem as a multi-period network design optimization problem, list the input parameters, key decision variables and the choice of objective functions. We also list the major assumptions that are used when building the model for CCUS multi-period infrastructure planning. Given a multi-period planning horizon and a finite budget allocation for each period, our goal is to make decisions regarding the following questions: Where to capture CO_2? When to start the capture process? What should the pipeline topology be? When should the pipelines be operational? Where to store the captured CO_2? How much to store in a sink over time?

Let $\mathcal{I}$ be the set of sources of stationary or point CO_2 emissions. For each source i, its geospatial coordinates given by latitude Lat_i and longitude Lon_i, and the annual emission amount E_i are known. Let $\mathcal{K}$

be the set of sinks for CO_2. We consider two types of sinks, namely the sequestration sinks and the utilization sinks. Let $\mathcal{K}_{seq}$ represent the set of sequestration sinks and let $\mathcal{K}_{util}$ represent the set of utilization sinks. For each sink k, its geospatial coordinates are given by latitude Lat_k and longitude Lon_k and for all sinks. The estimates of their storage volumes (or CO_2 retention amounts) are realized at different values with different probabilities. These realizations are available using the cumulative distribution functions (CDFs) provided by past geological surveys.

Let the planning horizon be divided into a set of $|\mathcal{T}|$ time periods. All time periods are considered to be of equal durations. A budget is allocated for each time period. Let the Ω be the set of scenario realizations. Every scenario is characterized by a different set of realizations of the storage volumes for each sink.

Given this information of sources, sinks, time periods considered, and scenario realizations, the objective is to perform optimal and timely selection of sources, sinks and determine the pipelines connecting them considering the allocated budget. Overall, we summarize the multi-period CCUS infrastructure planning problem as follows:
Given:

- $\mathcal{I}$ sources with their locations, emission amounts, types
- $\mathcal{K}_{seq}$ sequestration sites with their locations and CDFs
- $\mathcal{K}_{util}$ utilization sites with their locations and CDFs
- $\mathcal{T}$ time periods
- Ω scenarios with different realizations of the storage volumes
- Annual budget allocated for each time period
- Economic data for sizing of the capture plants to be built (see more details in Section 1 of the Supplementary Information)
- Distances between the sources and sinks, and appropriate pipeline costing factor (see more details in Section 2 of the Supplementary Information), and
- Planning horizon in which these decisions need to be made

Determine:

- All selected sources where capture plants need to be built
- Operational start time of the capture plants at the selected sources
- All selected sequestration and utilization sites where CO_2 will be sequestered or utilized
- Pipeline network topology with connectivity between the selected sources and sinks
- Pipeline transportation capacity and operation start time, and
- Amount of captured CO_2 to be transported from each selected source to each selected sink from beginning of some time period in some scenario realization of the storage volumes

The goal is to find the decisions that need to be taken for multi-period CCUS infrastructure deployment to maximize the total capture of CO_2 emissions in a way that the overall annual costs do not exceed the annual budget allocation.

We make the following assumptions:

1. The overall costs include: (1) fixed, investment and operating costs of CO_2 capture; (2) costs of CO_2 transportation; (3) cost of CO_2 storage over the operational life of a capture plant.
2. All infrastructure deployments are considered to be turnkey projects.
3. The capture plants are built at the scale equal to the annual emissions at the source.
4. Once a decision is made to transport some quantity CO_2 from a source to a sink, we assume that the same amount of CO_2 will be transported each year.
5. We assume a pipeline connects a single source to a single sink.
6. We assume a constant value of the CO_2 storage cost on a \$/ton basis which includes site screening, characterization, permitting and construction costs, its operational costs and post-injection site care costs (Smith et al., 2021).
7. All cash flows occurring in the future are adjusted to their Net Present Value (NPV) considering a discount factor.

3. Model formulations

3.1. Deterministic problem

One of the major decisions is to select where and when to establish CO_2 capture plants. To decide this, we define binary variable $\mu_{i,t}$ as follows:

$$\mu_{i,t} = \begin{cases} 1 & \text{if source } i \text{ is selected for } CO_2 \text{ capture and is} \\ & \quad \text{operational time period } t \text{ onward,} \\ 0 & \text{otherwise.} \end{cases}$$

We also define variable $h_{i,k,t}$ to represent the size or capacity of the pipeline to transfer captured CO_2 from source i to sink k starting at time period t. Let $y_{i,k,t}$ be the actual annual amount of CO_2 transported from source i to sink k starting from time period t.

With these key decision variables defined, we formulate the following MILP model. We call the problem PD, where D stands for deterministic CCUS network design and deployment optimization:

$$\begin{aligned} PD: \max \quad & \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CS_t E_i \mu_{i,t} - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} (CI_{i,t} + CO_{i,t})\mu_{i,t} \\ & - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CT_{i,k,t} h_{i,k,t} \\ & - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t} + \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t} \end{aligned} \tag{1}$$

s.t.

$$\sum_{i\in\mathcal{I}} (CI_{i,t} + CO_{i,t})\mu_{i,t} + \sum_{i\in\mathcal{I}} CT_{i,k,t} h_{i,k,t} + \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t} \le B_t, \qquad t \in \mathcal{T} \tag{2}$$

$$\sum_{k\in\mathcal{K}} y_{i,k,t} = E_i \mu_{i,t}, \qquad i \in \mathcal{I}, t \in \mathcal{T} \tag{3}$$

$$\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} y_{i,k,t} \le \frac{St_k}{L}, \qquad k \in \mathcal{K} \tag{4}$$

$$y_{i,k,t} \le h_{i,k,t}, \qquad i \in \mathcal{I}, k \in \mathcal{K}, t \in \mathcal{T} \tag{5}$$

$$\sum_{t\in\mathcal{T}} \mu_{i,t} \le 1, \qquad i \in \mathcal{I} \tag{6}$$

The objective (Eq. (1)) is to maximize the overall economic gains or benefits that could be achieved due to avoided social costs through CCUS over the entire planning horizon, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CS_t E_i \mu_{i,t}$, plus the net present value (NPV) of the revenue generated from CO_2-EOR activities, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t}$, minus the cash flows associated with CCUS infrastructure deployment and operations which include the NPV of the investment and operational cash flows, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}(CI_{i,t} + CO_{i,t})\mu_{i,t}$, network topology construction cash flows, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CT_{i,k,t} h_{i,k,t}$, and cash flows from for the storage operations, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t}$. Eq. (2) ensures that the cash flows associated with infrastructure deployments do not exceed the allocated budget for CCUS activities at any point in time. These cash flows refer to the turnkey values of a CO_2 project and include the initial investment cost and operational and storage costs over the entire life of the CCUS project. Eq. (3) ensures that if a capture plant is built at a source location, then all of the captured CO_2 (which is equal to the emission amount) must be transported to the sinks. Eq. (4) is the storage capacity constraint that limits the cumulative amount of CO_2 from all capture plants over their operational lives to not exceed the total storage capacity of a sink. Eq. (5) limits the amount transported through a pipeline by its capacity. Finally, Eq. (6) ensures that a plant starts operation at any source only at a single point in time over the entire planning horizon.

Solving the optimization problem PD provides us with the optimal deployment schedule for the entire horizon. Due to large benefits, it is

anticipated that the solution will entail nearly using all of the allocated budget in every time period. Note that PD is deterministic as the parameters are assumed to be known apriori. For example, the mean values for the storage volumes of the sequestration and utilization sites have been used.

3.2. Stochastic problem

Planning CCUS networks under uncertain geological storage volume setting may lead to undesirable deviations from the planned reductions in cumulative emissions. To avoid such deviations, we may need to modify the network at a later point in time, which may conflict with the decisions under the allocated budget for that time period. Thus, we need to find a design and its deployment schedule in a way that minimizes such undesirable deviations in all possible realizations of the underground storage volumes.

To that effect, we now introduce the set of scenario realizations Ω, with each individual scenario ω, having its own probability of occurrence p_ω. Specifically, each sink k has a different realization of its storage volume $D_{k,\omega}$ in every individual scenario ω. Let $y_{i,k,t,\omega}$ be the annual amount of CO_2 transported from source i to sink k from beginning of time period t in scenario ω. Depending on the realization of storage volumes for each sink k, denoted by $St_{k,\omega}$, it is undesirable if we cannot store what was originally planned to be captured. Let $xsl_{i,t,\omega}$ be the annual undesirable emission to the atmosphere from source i averaged over the entire operational life of the plant. In other words, $y_{i,k,t,\omega}$ and $xsl_{i,t,\omega}$ are the recourse actions to be taken based on scenario realization.

We now formulate the following stochastic MILP model for CCUS network design and multi-period deployment optimization under geological uncertainty;

$$\begin{aligned} PS: \max \quad & \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CS_t E_i \mu_{i,t} - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} (CI_{i,t} - CO_{i,t})\mu_{i,t} \\ & - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CT_{i,k,t} h_{i,k,t} \\ & - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{\omega\in\Omega} p_\omega \left[\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} + CS_t xsl_{i,t,\omega} \right. \\ & \left. - \sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t,\omega} \right] \end{aligned} \tag{7}$$

s.t.

$$\sum_{i\in\mathcal{I}} (CI_{i,t} + CO_{i,t})\mu_{i,t} + \sum_{i\in\mathcal{I}} CT_{i,k,t} h_{i,k,t} + \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} \le B_t, \qquad t\in\mathcal{T}, \omega\in\Omega \tag{8}$$

$$\sum_{k\in\mathcal{K}} y_{i,k,t,\omega} + xsl_{i,t,\omega} = E_i \mu_{i,t}, \qquad i\in\mathcal{I}, t\in\mathcal{T}, \omega\in\Omega \tag{9}$$

$$\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} y_{i,k,t,\omega} \le \frac{St_{k,\omega}}{L}, \qquad k\in\mathcal{K}, \omega\in\Omega \tag{10}$$

$$y_{i,k,t,\omega} \le h_{i,k,t}, \qquad i\in\mathcal{I}, k\in\mathcal{K}, t\in\mathcal{T}, \omega\in\Omega \tag{11}$$

$$\sum_{t\in\mathcal{T}} \mu_{i,t} \le 1, \qquad i\in\mathcal{I} \tag{12}$$

Again the objective, Eq. (7), is to maximize the overall economic benefits of CCUS through avoided social costs associated with CO_2 emissions over the entire planning horizon, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CS_t E_i \mu_{i,t}$, minus the cash flows associated with CCUS infrastructure deployment and operations which include the NPV of the investment and operational cash flows, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} (CI_{i,t} + CO_{i,t})\mu_{i,t}$, and the network topology deployment cash flows, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CT_{i,k,t} h_{i,k,t}$, minus the expected NPV of the cash flows associated with the CO_2 storage, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{\omega\in\Omega} p_\omega \sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega}$ and expected undesirable deviations from the planned capture amount, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{\omega\in\Omega} p_\omega \sum_{k\in\mathcal{K}} CS_t xsl_{i,t,\omega}$ plus the expected net present value (NPV) of the revenue generated from CO_2-EOR activities, $\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{\omega\in\Omega} p_\omega \sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t,\omega}$. Eq. (8) now ensures that the cash flows associated with infrastructure deployments, which includes costs associated with expected storage do not exceed the allocated budget for CCUS activities at any point in time in any scenario. Eq. (9) now enforces the mass balance constraint to be upheld for all scenarios. Likewise, for all scenarios, Eqs. (10) and (11) enforce the sinks' storage capacity constraints and the pipeline capacity constraints respectively.

Each scenario has a different realization of the underground storage volumes. For each sink, the CDFs of the storage volumes are available, leading to an infinite number of possible scenario realizations. To that end, we use the sample average approximation (SAA) method to generate an approximate problem $PSSAA$ with N independent and identically distributed (iid) scenarios. The scenarios are generated using the inverse transform sampling (ITS) method to capture the underlying probability distributions based on the CDFs as reported by the USGS (Warwick et al., 2013, 2019). For a detailed description of the ITS method, we direct the reader to Section 4.2 of Aras et al. (2025). We solve $PSSAA$ for this reduced set of scenarios, Ω_N to find an approximate solution to the true problem PS with infinite scenarios.

$$\begin{aligned} PSSAA: \max \quad & \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CS_t E_i \mu_{i,t} - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} (CI_{i,t} - CO_{i,t})\mu_{i,t} \\ & - \sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} CT_{i,k,t} h_{i,k,t} \\ & - \frac{1}{|\Omega_N|}\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}}\sum_{\omega\in\Omega_N} \left[\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} + CS_t xsl_{i,t,\omega} \right. \\ & \left. - \sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t,\omega} \right] \end{aligned} \tag{13}$$

s.t.

$$\sum_{i\in\mathcal{I}} (CI_{i,t} + CO_{i,t})\mu_{i,t} + \sum_{i\in\mathcal{I}} CT_{i,k,t} h_{i,k,t} + \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} \le B_t, \qquad t\in\mathcal{T}, \omega\in\Omega_N \tag{14}$$

$$\sum_{k\in\mathcal{K}} y_{i,k,t,\omega} + xsl_{i,t,\omega} = E_i \mu_{i,t}, \qquad i\in\mathcal{I}, t\in\mathcal{T}, \omega\in\Omega_N \tag{15}$$

$$\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} y_{i,k,t,\omega} \le \frac{St_{k,\omega}}{L}, \qquad k\in\mathcal{K}, \omega\in\Omega_N \tag{16}$$

$$y_{i,k,t,\omega} \le h_{i,k,t}, \qquad i\in\mathcal{I}, k\in\mathcal{K}, t\in\mathcal{T}, \omega\in\Omega_N \tag{17}$$

$$\sum_{t\in\mathcal{T}} \mu_{i,t} \le 1, \qquad i\in\mathcal{I} \tag{18}$$

Since the scenarios in Ω_N are i.i.d, each of them is equiprobable which is reflected in the expectation term of the objective function Eq. (13). Eqs. (14)–(17) have the same meaning as Eqs. (8)–(11) with the difference being that they are enforced for scenarios in Ω_N in the SAA problem instead of the original infinite set of scenarios, Ω. Eq. (18) remains the same as Eq. (12) as it is scenario independent.

4. Solution strategy

4.1. Deterministic planning using a rolling horizon approach

For a given geographical region with $|\mathcal{I}|$ sources and $|\mathcal{K}|$ sinks, with the entire horizon length being $|\mathcal{T}|$ time periods, solving the problem PD becomes difficult as the number of binary variables $\mu_{i,t}$ and the number of continuous variables $h_{i,k,t}$ and $y_{i,k,t}$ proportionately increase with the number of time periods considered. To address this, we employ a rolling horizon approach (Sethi and Sorger, 1991; Chand et al., 2002; Glomb et al., 2022) to find good planning schedules for CCUS infrastructure deployment.

In a rolling horizon approach, instead of finding the optimal network and deployment schedule for the entire horizon, ($t = 1, 2, \ldots |\mathcal{T}|$), at once, we consider a shorter horizon looking forward $|\mathcal{F}|$ time periods

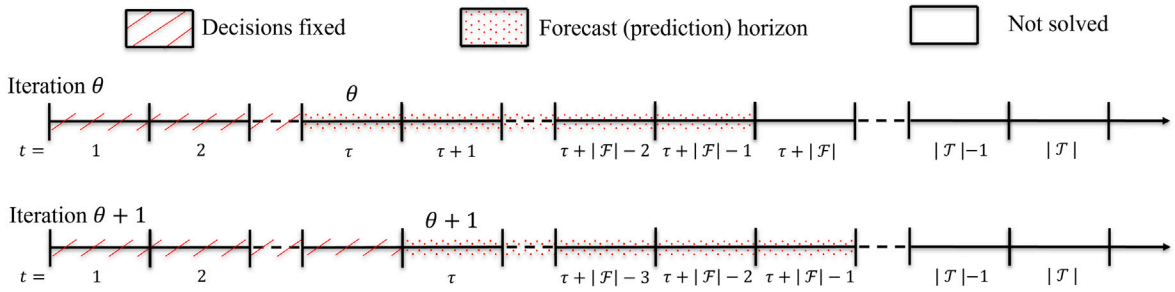


Fig. 2. Schematic representation of the rolling horizon approach. τ is the start of the forecast horizon up to which all decisions are fixed. At every iteration, the decisions for the time period τ are fixed.

into the future. Let this set of time periods be called the forecast or prediction time periods. To that effect, we consider problem $PD_{\tau,\mathcal{F}}$ that considers only the set of forecast time periods ($t = \tau, \tau+1, \dots \Gamma$). Here, τ is the beginning of the forecast horizon of current problem $PD_{\tau,\mathcal{F}}$, and $\Gamma = \min(\tau + |\mathcal{F}| - 1, |\mathcal{T}|)$. We solve $PD_{\tau,\mathcal{F}}$ and fix the deployment decisions for time τ. Then, we roll the forecast horizon window by one time period ($\tau = \tau + 1$) and solve model $PD_{\tau+1,\mathcal{F}}$ to find the deployment decisions for the next time period. A schematic of the rolling horizon procedure is presented in Fig. 2. Let $\mathcal{I}_{\mathcal{A}}$ be the set of all sources where CO_2 capture plants have not been built before the beginning of the current forecast window and let $\mathcal{I}_D$ be the set of all sources where CO_2 capture plants have been built before the beginning of the current forecast window. Before the start of the entire planning horizon, no capture plants have been built; $\mathcal{I}_{\mathcal{A}} = \mathcal{I}$ and $\mathcal{I}_D = \{\}$. At every iteration, the sources where the capture plants have been built at time τ are removed from the set $\mathcal{I}_{\mathcal{A}}$ and added to the set $\mathcal{I}_D$. Solving this model for a certain forecast window also entails making the operational decisions of the amount of CO_2 to be transported to the sinks from the beginning of each time period in that window. These include $y_{i,k,\tau}$ or the operational decisions for the current time period which are now fixed. The sink volumes are updated for the next forecast window by subtracting the amount of cumulative CO_2 transported from all sources to a sink beginning at current time period τ from the sink volumes that were available for the current forecast window. Let the available storage volume of a sink k at the beginning of some time period τ be denoted by St_k^{τ}. Then, we roll the forecast horizon window by one time period ($\tau = \tau + 1$) and solve $PD_{t,\mathcal{F}}$ to find the deployment decisions for the next time period. The process is repeated iteratively until the decisions for all time periods in the entire horizon have been fixed.

The rolling horizon approach allows us to obtain a good deployment schedule over the entire horizon. At every iteration, we consider the optimal solution vector of the deployment decisions, $[\mu^*_{i,\tau}, h^*_{i,k,\tau}]$, to be denoted as x^*_{τ}. When $PD_{\tau,\mathcal{F}}$ is iteratively solved for all periods, all vectors are aggregated to obtain a single deployment schedule for the entire horizon. We denote this solution by $X^* = [x^*_1, x^*_2, \dots, x^*_{|\mathcal{T}|}]$. Since the storage volumes are updated at every iteration, this aggregated schedule will not lead to any violations of the storage capacity constraints. This rolling horizon procedure is formally defined in Algorithm 1. Since the entire horizon is not included in the forecast set at once ($|\mathcal{F}| < |\mathcal{T}|$), there exists a tradeoff between the solution quality and computational complexity and the schedule obtained may not be globally optimal.

$$
\begin{aligned}
PD_{\tau,\mathcal{F}}: \max \quad & \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma} CS_t E_i \mu_{i,t} - \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma}(CI_{i,t} + CO_{i,t})\mu_{i,t} \\
& - \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma} CT_{i,k,t} h_{i,k,t} \\
& - \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t} + \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma}\sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t} \qquad (19) \\
\text{s.t.} \quad & \\
& \sum_{i\in\mathcal{I}_{\mathcal{A}}}(CI_{i,t} + CO_{i,t})\mu_{i,t} + \sum_{i\in\mathcal{I}_{\mathcal{A}}} CT_{i,k,t} h_{i,k,t} \\
& + \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t} \le B_t, \qquad t \in \Gamma \qquad (20) \\
& \sum_{k\in\mathcal{K}} y_{i,k,t} = E_i \mu_{i,t}, \qquad i \in \mathcal{I}_{\mathcal{A}}, t \in \Gamma \qquad (21) \\
& \sum_{i\in\mathcal{I}_{\mathcal{A}}}\sum_{t\in\Gamma} y_{i,k,t} \le \frac{St_k^{\tau}}{L}, \qquad k \in \mathcal{K} \qquad (22) \\
& y_{i,k,t} \le h_{i,k,t}, \qquad i \in \mathcal{I}_{\mathcal{A}}, k \in \mathcal{K}, t \in \Gamma \qquad (23) \\
& \sum_{t\in\Gamma} \mu_{i,t} \le 1, \qquad i \in \mathcal{I}_{\mathcal{A}} \qquad (24)
\end{aligned}
$$

Eqs. (1)–(6) and (19)–(24) have equivalent interpretations with the only difference being that the first set of equations considering the entire horizon and the second set considering the decisions that pertain only to the considered forecast time periods.

Algorithm 1 Rolling horizon approach for finding CCUS infrastructure deployment schedules under a deterministic storage volume setting.

1: $I_A \leftarrow I$, $I_D \leftarrow \emptyset$, $\tau \leftarrow 1$, $St_k^1 \leftarrow St_k$, ▷ Initialization
2: **while** $\tau \le |\mathcal{T}|$ **do**
3: $\hat{\boldsymbol{\mu}}_{i,t}, \hat{\boldsymbol{h}}_{i,k,t}, \hat{\boldsymbol{y}}_{i,k,t} = \arg\min PD_{\tau,\mathcal{F}}$ ▷ Find deployment and operational decisions
4: $\hat{\boldsymbol{x}}_{\tau} = [\hat{\boldsymbol{\mu}}_{i,\tau}, \hat{\boldsymbol{h}}_{i,k,\tau}]$ ▷ Fix decisions obtained (schedule) for time period τ
5: **for** $i = 1, 2, ..., |\mathcal{I}|$ **do**
6: **if** $\hat{\mu}_{i,\tau} = 1$ **then** ▷ Check if plant is built in current iteration
7: $I_D = I_D \cup \{i\}$
8: $I_A = I_A - \{i\}$ ▷ Remove from set of available sources
9: **end if**
10: **end for**
11: **for** $k = 1, 2, ..., |\mathcal{K}|$ **do**
12: $St_k^{\tau+1} = St_k^{\tau} - L \sum_{i\in\mathcal{I}_{\mathcal{A}}} \hat{y}_{i,k,\tau}$ ▷ Update storage volume available for every sink
13: **end for**
14: $\tau = \tau + 1$
15: **end while**
16: $\hat{\boldsymbol{X}} = [\hat{\boldsymbol{x}}_{\boldsymbol{1}}, \hat{\boldsymbol{x}}_{\boldsymbol{2}}, ...\hat{\boldsymbol{x}}_{|\mathcal{T}|}]$ ▷ Aggregate schedule

4.2. Stochastic planning using a rolling horizon approach

Solving problem $PS-SAA$ is even more difficult than solving problem PD as the number of continuous variables $y_{i,k,t,\omega}$ and $xsl_{i,k,t,\omega}$ also depends upon the number of scenario realizations considered, or $|\Omega|$. In this section, we present the rolling horizon approach to find good planning schedules for CCUS infrastructure deployment under an uncertain geological storage volume setting. We now introduce model $PSSAA_{\tau,\mathcal{F}}$ which include the set of i.i.d scenarios, with each scenario having a different realization of the storage volumes. We solve model

$PSSAA_{\tau,\mathcal{F}}$ to find and fix the deployment decisions for time τ, $\mu_{i,\tau}$ and $h_{i,k,\tau}$, which are scenario independent. We then roll the forecast horizon window by one time period to find and fix the deployment decisions for the next time period. At every iteration, the set of available sources is updated. Since the annual amount of CO_2 transported to the sinks may differ across scenarios, the storage volumes need to be updated accordingly for each scenario. Similar to the deterministic approach, let the optimal solution vector of the deployment decisions to be fixed, $[\hat{\mu}_{i,\tau}, \hat{h}_{i,k,\tau}]$ be denoted as $\hat{x}_\tau$. Again, we solve $PSSAA_{\tau,\mathcal{F}}$ iteratively until the solution vectors for all time periods are obtained which are then aggregated to obtain $\hat{X}_N$ where the subscript N stands for the reduced set of N scenarios in $PSSAA$. This rolling horizon procedure is formally defined in Algorithm 2.

$$
\begin{aligned}
PSSAA_{\tau,\mathcal{F}}: \max \quad & \sum_{i\in\mathcal{I}_\mathcal{A}}\sum_{t\in\Gamma} CC_t^{Soc} E_i \mu_{i,t} - \sum_{i\in\mathcal{I}_\mathcal{A}}\sum_{t\in\Gamma}(CI_{i,t}+CO_{i,t})\mu_{i,t} \\
& - \sum_{i\in\mathcal{I}_\mathcal{A}}\sum_{t\in\Gamma} CT_{i,k,t} h_{i,k,t} \\
& - \sum_{i\in\mathcal{I}_\mathcal{A}}\sum_{t\in\Gamma}\sum_{\omega\in\Omega} p_\omega \Bigg[\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} + CS_t xsl_{i,t,\omega} \\
& - \sum_{k\in\mathcal{K}_{util}} R_t y_{i,k,t,\omega} \Bigg]
\end{aligned} \tag{25}
$$

s.t.

$$
\sum_{i\in\mathcal{I}_\mathcal{A}}(CI_{i,t}+CO_{i,t})\mu_{i,t} + \sum_{i\in\mathcal{I}_\mathcal{A}} CT_{i,k,t} h_{i,k,t} + \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}} CSt_t y_{i,k,t,\omega} \le B_t, \qquad t\in\Gamma, \omega\in\Omega \tag{26}
$$

$$
\sum_{k\in\mathcal{K}} y_{i,k,t,\omega} + xsl_{i,t,\omega} = E_i \mu_{i,t}, \qquad i\in\mathcal{I}_\mathcal{A}, t\in\Gamma, \omega\in\Omega \tag{27}
$$

$$
\sum_{i\in\mathcal{I}_\mathcal{A}}\sum_{t\in\Gamma} y_{i,k,t,\omega} \le \frac{St^\tau_{k,\omega}}{L}, \qquad k\in\mathcal{K}, \omega\in\Omega \tag{28}
$$

$$
y_{i,k,t,\omega} \le h_{i,k,t}, \qquad i\in\mathcal{I}_\mathcal{A}, k\in\mathcal{K}, t\in\Gamma, \omega\in\Omega \tag{29}
$$

$$
\sum_{i\in\Gamma}\mu_{i,t} \le 1, \qquad i\in\mathcal{I}_\mathcal{A} \tag{30}
$$

Eqs. (13)–(18) and (25)–(30) have equivalent interpretations with the only difference being that the first set of equations considering the entire horizon and the second set considering the decisions that pertain only to the considered forecast time periods.

Now that an aggregated schedule for CCUS infrastructure deployment under uncertain volume setting is obtained, we fix the identified deployment decisions $(\hat{\mu}_{i,t}, \hat{h}_{i,k,t})$ for every time period in the model $PSSAA$. The makes $PSSAA$ a linear program which can be solved to get the values of the total economic gain due to avoided emissions, and the total undesired deviation from the planned capture amount for that particular schedule (solution vector). This also allows the values of the total CO_2 annual emissions for each period to be obtained. Note that the deployment decisions in the aggregated schedule $(\hat{\mu}_{i,t}, \hat{h}_{i,k,t})$ are obtained from solving $PSSAA_{\tau,\mathcal{F}}$ which ensures the feasibility of the budget constraint across all time periods.

Let the optimal objective value of PS be $\mathbb{E}\left[f(X^*,\Omega)\right]$, or $Z^\star$. As it is not possible to compute this true optimal objective value, we can generate valid lower bounds and upper bounds using statistics (Ntaimo, 2024). Since $\hat{X}_N$ is a feasible solution to the approximate problem, Eq. (31) always holds for any set of samples generated.

$$
\frac{1}{|\Omega_N|}\sum_{\omega=1}^{|\Omega_N|} f(\hat{X}_N,\Omega_N) \ge \frac{1}{|\Omega_N|}\sum_{\omega=1}^{|\Omega_N|} f(X^*,\Omega_N) \tag{31}
$$

Algorithm 2 Rolling horizon approach for finding CCUS infrastructure deployment schedules under uncertain storage volume setting.

$I_A \leftarrow I$, $I_D \leftarrow \emptyset$, $\tau \leftarrow 1$, $St^1_k \leftarrow St_k$, ▷ Initialization
while $\tau \le |\mathcal{T}|$ **do**
  $\hat{\boldsymbol{\mu}}_{i,t}, \hat{\boldsymbol{h}}_{i,k,t}, \hat{\boldsymbol{y}}_{i,k,t,\omega} = \arg\min PS - SAA_{\tau,\mathcal{F}}$ ▷ Find deployment and operational decisions
  $\hat{\boldsymbol{x}}_\tau = [\hat{\boldsymbol{\mu}}_{i,\tau}, \hat{\boldsymbol{h}}_{i,k,\tau}]$ ▷ Fix decisions obtained (schedule) for time period τ
  for $i = 1, 2, ..., |\mathcal{I}|$ **do**
    if $\hat{\mu}_{i,\tau} = 1$ **then** ▷ Check if plant is built in current iteration
      $I_D = I_D \cup \{i\}$
      $I_A = I_A - \{i\}$ ▷ Remove from set of available sources
    end if
  end for
  for $k = 1, 2, ..., |\mathcal{K}|$ **do**
    for $\omega = 1, 2, ..., |\Omega|$ **do** ▷ For every scenario,
      $St^{\tau+1}_{k,\omega} = St^\tau_{k,\omega} - L\sum_{i\in\mathcal{I}_\mathcal{A}} \hat{y}_{i,k,\tau,\omega}$ ▷ Update storage volume available for every sink
    end for
  end for
  $\tau = \tau + 1$
end while
$\hat{\boldsymbol{X}}_N = [\hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, ...\hat{\boldsymbol{x}}_{|\mathcal{T}|}]$ ▷ Aggregate schedule

$$
\mathbb{E}\left[\frac{1}{|\Omega_N|}\sum_{\omega=1}^{|\Omega_N|} f(\hat{X}_N,\Omega_N)\right] \ge \mathbb{E}\left[\frac{1}{|\Omega_N|}\sum_{\omega=1}^{|\Omega_N|} f(X^*,\Omega_N)\right] \tag{32}
$$

We get Eq. (32) using the expectation operator on both sides, and as $N\to\infty$, $\mathbb{E}[\hat{Z}_N] \ge \mathbb{E}[f(X^*,\Omega)]$. Hence, $\mathbb{E}[\hat{Z}_N]$ is a valid upper bound on the true optimal. To estimate the upper bound $\mathbb{E}[\hat{Z}_N]$, we generate M i.i.d batches (sets of scenarios) of N samples each and solve the approximated problem $PSSAA_{\tau,\mathcal{F}}$ for every time period and aggregate the solution for each batch. Let U_{MN} be the upper bound on the true optimal given by Eq. (33).

$$
U_{MN} = \frac{1}{M}\sum_{j=1}^{M} \hat{Z}^j_N \tag{33}
$$

Given any feasible solution vector, $\hat{X}_N$, Eq. (34) always holds:

$$
\mathbb{E}[f(X^*,\Omega)] \ge \mathbb{E}[f(\hat{X}_N,\Omega)] \tag{34}
$$

To compute the lower bound, $\mathbb{E}[f(\hat{X}_N,\Omega)]$, a very large number of scenarios, $\overline{N}$, are generated. The solution vector $\hat{X}_N$ is then fixed into each of these scenarios which are considered individually to solve problem $PSSAA$ as an LP. Let the solution of this LP for some scenario n in $\overline{N}$ be denoted by $\hat{Z}^n$. The lower bound to the true optimal objective value is given by Eq. (35)

$$
L_{\overline{N}} = \frac{1}{\overline{N}}\sum_{j=1}^{\overline{N}} \hat{Z}^j \tag{35}
$$

Since we have considered M i.i.d batches resulting in M good candidate solutions or CCUS deployment schedules, to choose the best solution vector among those, we follow the procedure in Kall and Mayer (Kall and Mayer, 2005) to find the best deployment schedule with the lowest duality gap which fits best in all scenarios considered across all batches in terms of the total economic gain.

5. Results

We apply the methodology to the region around the Illinois Basin which includes 3 storage sites and 3 utilization sites in this basin as reported in the GHGRP and USGS databases. Two additional utilization sites form the Appalachian Basin are included for our case study. For

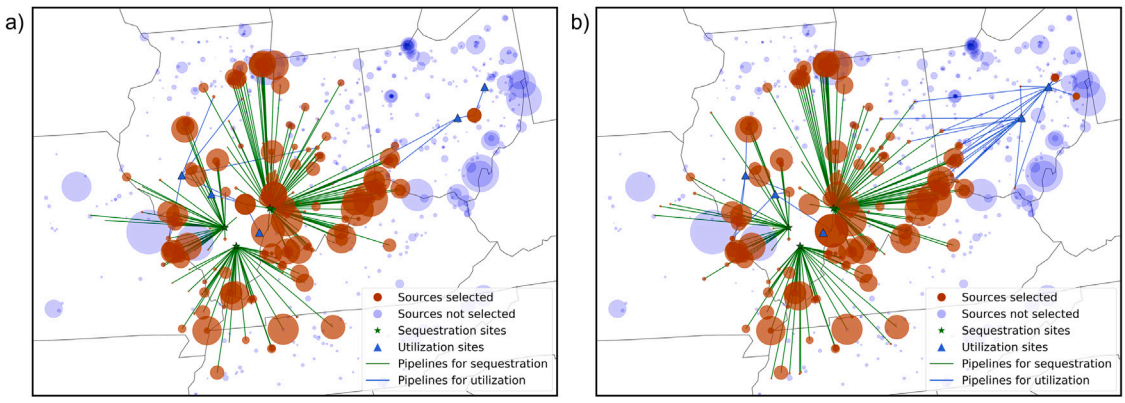


Fig. 3. Final configuration of the CCUS networks in the Illinois Basin region at the end of planning horizon: (a) Using a deterministic approach, (b) Using a stochastic approach.

the same region, we have 820 CO_2 emission sources located in the states of Illinois, Indiana, Ohio, Kentucky and Missouri with a total estimated CO_2 emissions of 458,673 kt/yr. The length of the total planning horizon is 25 years, representative of the years 2026–2050, and we consider a forecast horizon of 5 years. Over this 25 year period, we consider an annual budget allocation of $4.8 billion. We assume a discount factor of 3% and the social cost of emissions to be $42/ton of CO_2 emitted (EPA, 2016). In the SAA scheme, we consider 10 batches each with a set of 25 scenarios ($M = 10, N = 25$). $PSSAA_{\tau,\mathcal{F}}$ was solved in its deterministic equivalent form for each of the iterations in the rolling horizon algorithm. To compute the lower bound and to quantify the undesirable deviations, we have considered 5000 scenarios ($\overline{N} = 5000$) for the best vector obtained. All problems are solved in Python v3.9.6 using the solver Gurobi v11.0.3 to 1% optimality gap using a Linux OS with a 13th Gen Intel i7-13700 processor using 24 CPU cores with 32 GB RAM.

With a total planning horizon length of 25 years, from 2026 to 2050, we solve the optimization problem and Fig. 3a and b show the configuration of the complete network at the end of the planning horizon. The orange bubbles represent the sources selected for capture and the blue bubbles represent the ones that are not. The size of the bubbles corresponds to the amount of CO_2 captured/emitted. The sequestration and utilization pipelines are shown with green and blue lines, respectively. All 3 storage sites and all 5 utilization sites in the region are selected to manage the captured CO_2. More pipeline connections are present in the network designed using the stochastic approach than in the one designed using a deterministic approach as these extra connections can be attributed to the allowance of recourse actions in the stochastic approach.

Figs. 4 and 5 show the evolution of the CCUS network over time and the deployment decisions made every year obtained from solving $PD_{\tau,\mathcal{F}}$ and $PSSAA_{\tau,\mathcal{F}}$, respectively. In both cases, due to economies of scale coming into play, the construction of larger plants is prioritized to effectively manage CO_2 as it results in lower costs for every ton of CO_2 captured. In both cases, utilization of CO_2 for EOR is maximized as it generates revenue for the CCUS network. In Fig. 4, it can be seen that the connections from sources to utilization sites are made in the initial years only. In contrast, such connections are built even in the later stages of the entire planning horizon when there is uncertainty in the retention volumes. This allows for corrective actions if the utilization volumes are found to be insufficient in some scenario realizations, as can be seen in Fig. 5.

As the sources with larger emissions are accounted for in the earlier time periods, a uniform budget allocation leads to a reduction in the total management capacity of the capture facilities built every period over time. This can be seen in Fig. 6a, where capture plants totaling up to 10,596 kt/yr of CO_2 capture are constructed in the first time period and this total reduces to 9,641 kt/yr of total capture capacities of the plants built in the final time period. Fig. 6b shows the total emissions in the region starting from 458,673 kt/yr, when no plants have been constructed, which are reduced to 204,723 kt/yr by the end of the planning horizon. Thus, a budget allocation of $4.8 billion every year results in a 55.3% reduction in total annual emissions at the end of the planning horizon. These values have been reported for the network design and schedule obtained under an uncertain volume setting.

To compare the deterministic and stochastic network designs and deployment schedules in terms of managing CO_2, we fix each of the designs in 5000 scenarios and compute the undesirable deviations arising from failing to capture the planned amount. The network and deployment schedule obtained from using mean values of the storage volumes result in an average of 4,795 kt of deviations from the planned capture per scenario over the operational lives of all plants, whereas using a stochastic approach results in 145.24 kt of deviations for the planned capture. A 96.97% reduction in the undesirable deviations is seen and highlights the needs of a stochastic approach for designing CCUS networks and finding a deployment schedule for the same.

To study the effect of variation in the annual budget allocation on the topology of the network and the reductions in emissions, we also consider three more annual budget allocations: 50%, 75% and 125% of the annual base budget of $4.8 billion per year. Fig. 7 shows the final network topology at the end of the planning horizon for the region under study with these budget allocation conditions. Under all budget allocations, utilization is prioritized as it generates revenue and an increase in the allocated budget unsurprisingly leads to more sources selected for capture. If the budget allocated is not enough to build a single capture plant even at one of the largest emission sources, it may be suboptimal from a cumulative emissions reduction standpoint. This is evident from the 50% budget allocation case where some of the largest emission sources cannot be selected and these sources may be responsible for effective CO_2 capture on a $/ton basis due to economies of scale. Fig. 8 shows the reductions in emissions over time under different budget allocations, from a 27.5% reduction with an annual budget of $2.4 billion to a 64.2% reduction with an annual budget of $6 billion. This also suggests that an optimal tradeoff may exist between the allocated budget and the cost-effectiveness CO_2 captured on a $/ton basis.

6. Conclusions

This study presents an approach for designing CCUS networks and finding the optimal deployment schedule over a deployment horizon, while incorporating uncertainties in geological storage capacities. Our findings reveal that it is much more economically viable and environmentally favorable to build capture plants at sources with higher CO_2 emissions earlier. Furthermore, a multi-period approach provides us with a deployment schedule and indicates the times at which such

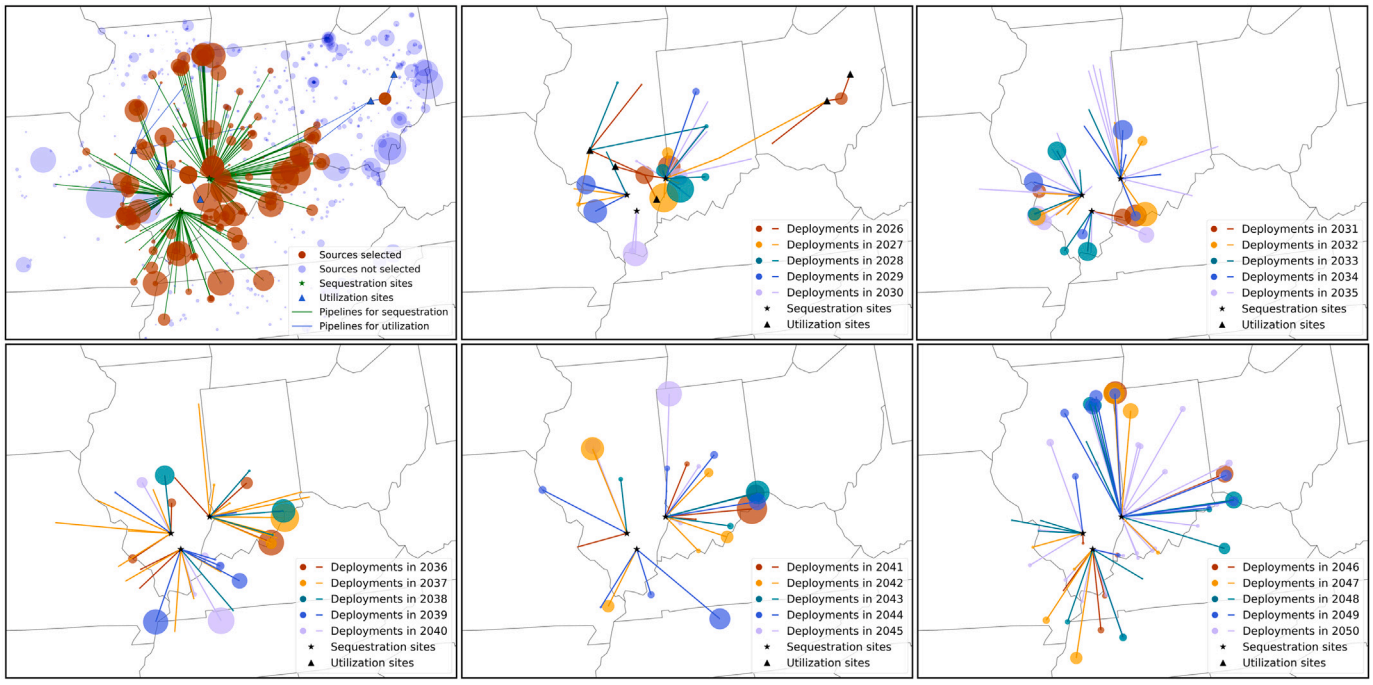


Fig. 4. Evolution of the network for the deterministic approach over time. Each image shows the decisions made in every 5 year period in the planning horizon.

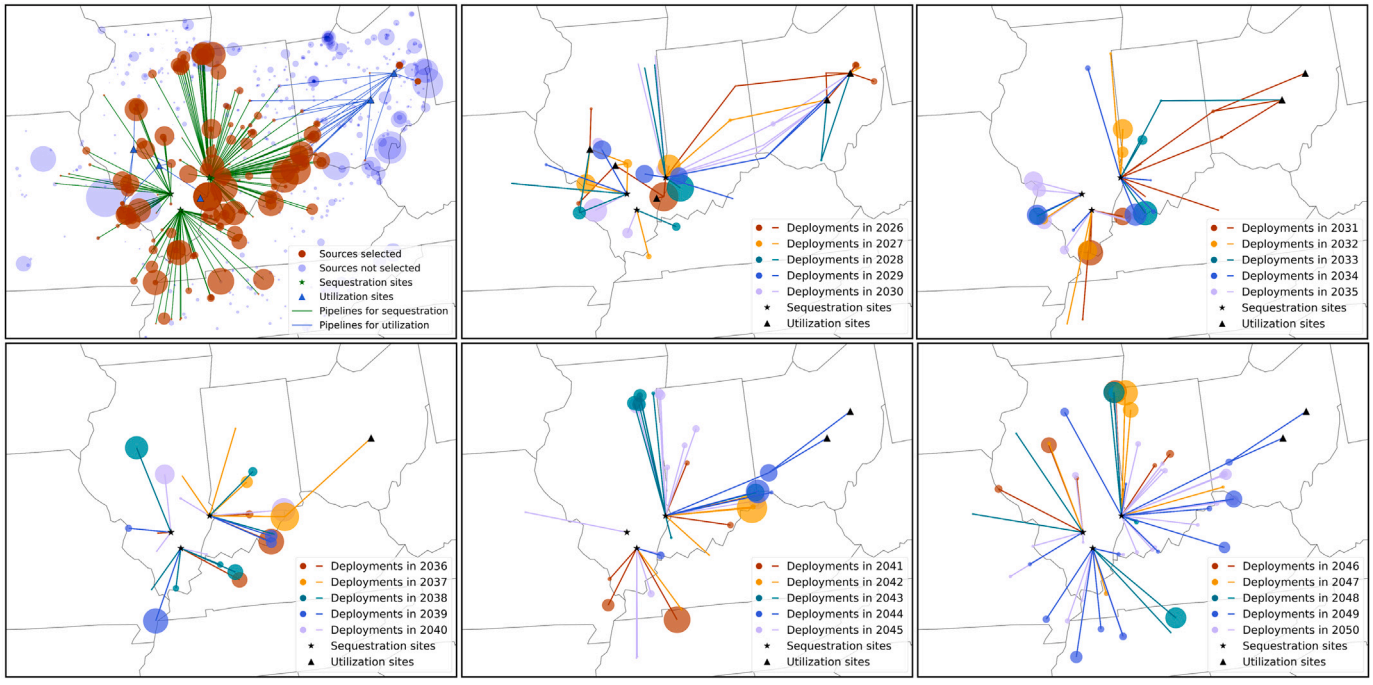


Fig. 5. Evolution of the network for the stochastic approach over time. Each image shows the decisions made in every 5 year period in the planning horizon.

deployments are the most impactful. Emphasizing CO_2 use in enhanced oil recovery introduces a strong financial driver into the network design process, leading to an early preference for sites which manage the captured CO_2 that offer economic incentives over ones that do not. At the same time, when uncertainty in the storage volumes is considered, allowing for recourse actions warrants for more pipeline connections over the network to minimize deviations from the planned capture amount. The extent of annual CO_2 emission reduction is directly tied to both the yearly commissioning of capture plants and the amount of budget allocation to the network. Finally, a stochastic approach for CCUS network design allows for a significant reduction in the expected deviations from the planned capture over a deterministic approach.

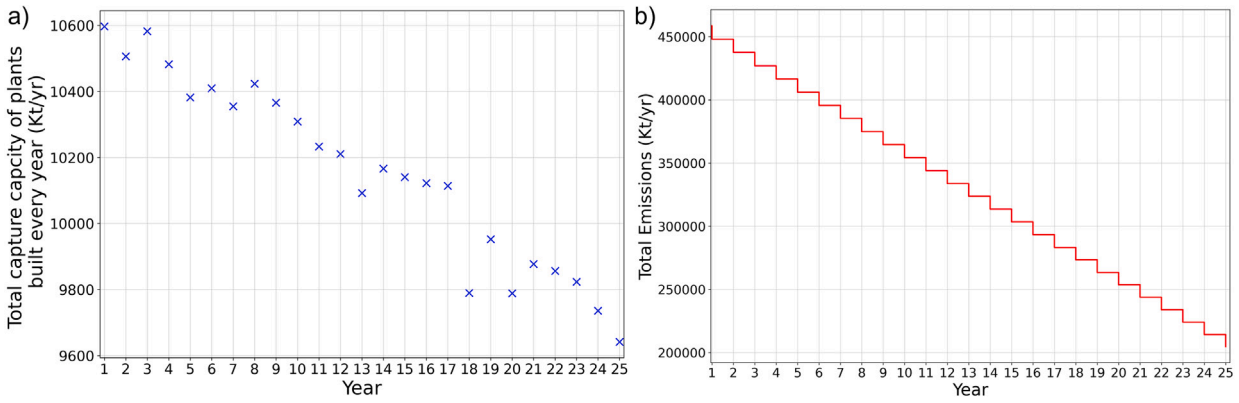


Fig. 6. Tracking total capture capacities of plants built every year and reduction in CO_2 emissions over time.

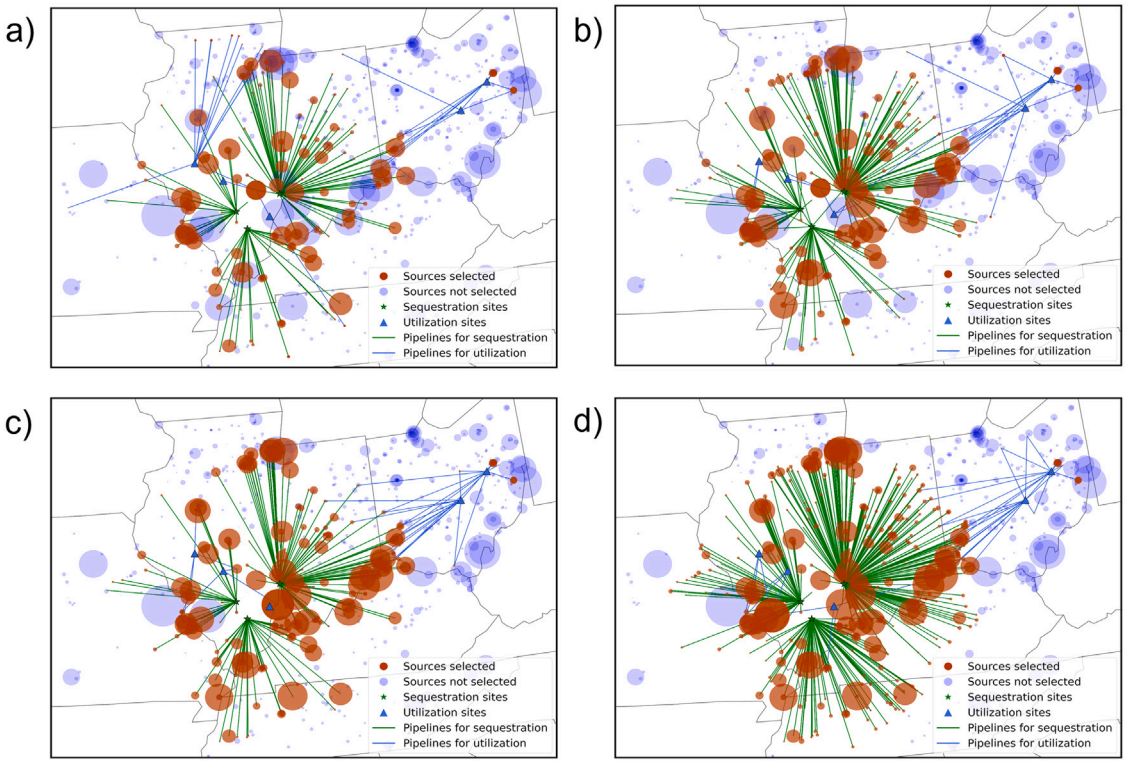


Fig. 7. Final configuration of the CCUS networks in the Illinois Basin region at the end of planning horizon using the stochastic approach: (a) 50% of base budget, (b) 75% of base budget, (c) base budget, and (d) 125% of base budget.

One should note that economic factors such as CO_2 selling prices and the social cost of CO_2 emissions may be entirely dynamic. Furthermore, ongoing innovation in capture and utilization technologies has the potential to significantly lower the cost of these processes, ultimately enabling more deployment activities than were originally planned for a given budget in the future. If data is provided on CO_2 emission sources and underground storage estimates, our CCUS multi-period infrastructure planning framework is broadly applicable to any region even though this study primarily focuses on the region around the Illinois Basin. By identifying where and when CCUS initiatives can have the largest impact, this approach equips decision-makers with the insights needed to allocate funding effectively and prioritize strategic investments. These results also strengthen the case for CCUS by accounting for geological storage volume uncertainty explicitly, by increasing confidence in large-scale investments and increases the viability of CCUS as a cornerstone of global decarbonization efforts.

7. Nomenclature

CCUS	Carbon Capture, Utilization, & Storage
CDF	Cumulative Distribution Function
EOR	Enhanced Oil Recovery
GHGRP	Greenhouse Gas Reporting Program
i.i.d	independent & identically distributed
LP	Linear Program
MILP	Mixed Integer Linear Program
MINLP	Mixed Integer Non-Linear Program
NPV	Net Present Value
SAA	Sample Average Approximation
SP	Stochastic Programming
USGS	United States Geological Survey

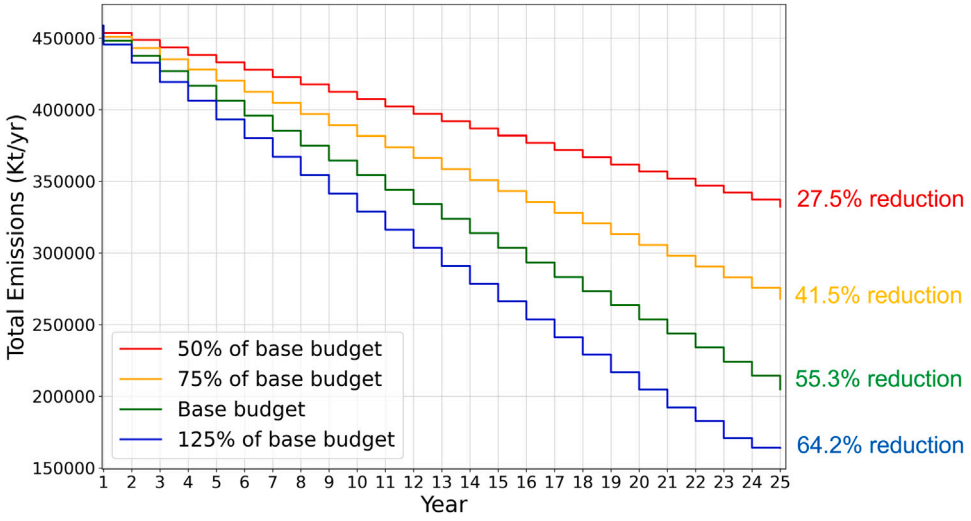


Fig. 8. Reduction in CO_2 emissions over time under different budget allocations.

8. List of symbols

Sets

$\mathcal{F}$	Set of forecast (prediction) time periods
$\mathcal{I}$	Set of sources
$\mathcal{K}$	Set of sites
$\mathcal{K}_{seq}$	Set of sequestration sites
$\mathcal{K}_{util}$	Set of utilization sites
$\mathcal{T}$	Set of time periods
Ω	Set of scenarios
Ω_N	Reduced set of scenarios

Parameters

$CI_{i,t}$	NPV of investment cost of capture plant built at source i at beginning of time period t
$CO_{i,t}$	Cumulative NPV of operational costs of capture plant built at source i at beginning of time period t
R_t	Cumulative NPV of revenue generated per kton of CO_2 utilized for L time periods from beginning of time period t
CS_t	Cumulative NPV of social costs incurred per kton of CO_2 emitted for L time periods from beginning of time period t
CSt_t	Cumulative NPV of storage costs per kton of CO_2 stored for L time periods from beginning of time period t
$CT_{i,k,t}$	NPV of pipeline built from source i to sink k at beginning of time period t
$dist_{i,k}$	Distance between source i and sink k
E_i	Annual emission at source i
L	Operational life of plant
St_k	Maximum available sequestration or utilization amount in site k
St_k^τ	Maximum available sequestration or utilization amount in site k at beginning of forecast window τ
$St_{k,\omega}$	Maximum yearly available sequestration or utilization amount in site k in scenario ω
$St_{k,\omega}^\tau$	Maximum yearly available sequestration or utilization amount in site k in scenario ω at beginning of forecast window τ

Continuous Variables

$h_{i,k,t}$	Transportation capacity of pipeline built from source i to sink k at beginning of time period t
$xsl_{i,t,\omega}$	Annual undesirable emission from capture plant i beginning at time period t occurring for L time periods
$y_{i,k,t}$	Annual amount of CO_2 transported from source i to sink k beginning at time period t for L time periods
$y_{i,k,t,\omega}$	Annual amount of CO_2 transported from source i to sink k beginning at time period t in scenario ω for L time periods

Binary Variable

$$\mu_i = \begin{cases} 1 & \text{if source } i \text{ is selected for capture and is operational} \\ & \quad \text{time period } t \text{ onward,} \\ 0 & \text{otherwise.} \end{cases}$$

CRediT authorship contribution statement

Chinmay M. Aras: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M.M. Faruque Hasan:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compchemeng.2025.109300.

Data availability

Data will be made available on request.

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